



DO OTHER ANIMALS HAVE MIRROR NEURONS?

Mirror neurons were first discovered in macaque monkeys. They have also been found in some birds, such as songbirds, and more recently in rats.

1 Watching a body movement

Watching a person perform an action not linked to an object, such as chewing, activates the premotor area in the observer. This is an area that is linked to rehearsing planned sequences of action. It also activates areas in the primary motor area associated with mouth and jaw movements.

ACTION WITH NO OBJECT

2 Watching action on an object

Watching an action directed at an object, such as a person biting into a fruit, activates similar areas of the motor cortex. However, mirror neurons also fire in an additional area, the parietal cortex, which is involved in interpreting sensory input as well as providing information about the body's position.

ACTION WITH AN OBJECT

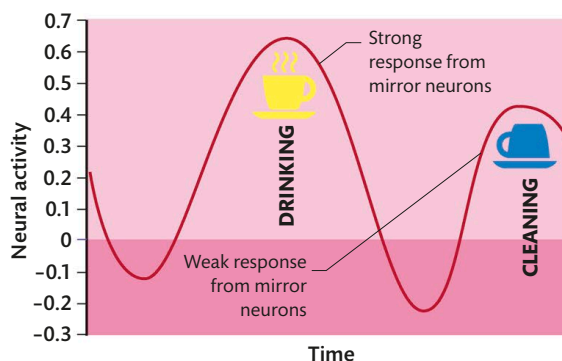
YAWNING

Mirror neurons may play a role in "contagious yawning"—the impulse to yawn when we see someone else yawning. fMRI scans of people who watched videos of someone else yawning showed activity in the right inferior frontal gyrus, an area associated with mirror neurons.



Understanding intention

Mirror neurons are activated in different ways when we see others performing particular actions, suggesting they could play a role in decoding intention. Watching similar actions observed in different contexts—such as watching someone pick up a cup either to drink from it or to clean it up—triggers different levels of neural activity in the inferior frontal gyrus; an area of the brain that directs our attention to objects in our environment.



Intention and brain activity

Activity in the brain is greater when a person watches someone lift a cup to drink rather than when they watch someone pick it up to clear it away. Some scientists suggest this may be because drinking has a greater biological function than cleaning.

THE BRAIN WAVES OF MUSICIANS COME INTO SYNC WHEN THEY PLAY TOGETHER

